

Crops

|              | Crop 1     | Crop 2  | Crop 3 | Crop 4  | Crop 5 | Crop 6 | Crop 7 | Crop 8     | Crop 9       | Crop 10 | Crop 11         |
|--------------|------------|---------|--------|---------|--------|--------|--------|------------|--------------|---------|-----------------|
|              | Horse Bean | Einkorn | Emmer  | Lentils | Peas   | Olives | Grapes | Grass Peas | Bitter Vetch | Barley  | FreeHull Barley |
| Yield        | 500        | 500     | 500    | 300     | 150    | 600    | 600    | 500        | 400          | 500     | 500             |
| Units        | Ha         | Ha      | Ha     | Ha      | Ha     | Ha     | Ha     | Ha         | Ha           | Ha      | Ha              |
| Mcals/kg     | 3          | 3       | 3      | 3       | 3.25   | 4      | 3      | 3          | 3            | 3       | 3               |
| Fallow Ratio | 1          | 1       | 1      | 1       | 1      | 0      | 0      | 1          | 1            | 1       | 1               |
| Fallow MCals | 500        | 500     | 500    | 500     | 500    | 500    | 500    | 500        | 500          | 500     | 500             |
| Fodder Kg    | 300        | 300     | 300    | 300     | 100    | 0      | 0      | 300        | 300          | 300     | 300             |
| Fodder MCals | 0.8        | 0.8     | 0.8    | 0.8     | 0.8    | 0.8    | 0.8    | 0.8        | 0.8          | 1.42    | 0.8             |

Animal Definitions

|                                      | Animal 1 | Animal 2 | Animal 3 | Animal 4 | Animal 5 | Animal 6 | Animal 7 | Animal 8 |
|--------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|
|                                      | Sheep    | Pig      | Cow      | Goat     |          |          |          |          |
| Mcalories Kg <sup>-1</sup>           | 3.200    | 3.200    | 3.200    | 3.200    |          |          |          |          |
| Usable %                             | 50.00%   | 50.00%   | 50.00%   | 50.00%   |          |          |          |          |
| Average Kill Weight                  | 30       | 100      | 400      | 30       |          |          |          |          |
| Weaning Weight                       | 8        | 2        | 50       | 8        |          |          |          |          |
| Adult Weight                         | 60       | 200      | 625      | 60       |          |          |          |          |
| Grow Time                            | 40       | 40       | 50       | 40       |          |          |          |          |
| Death Rate                           | 30.00%   | 30.00%   | 20.00%   | 30.00%   |          |          |          |          |
| Conception Efficiency                | 80.00%   | 70.00%   | 85.00%   | 80.00%   |          |          |          |          |
| Sex Maturity                         | 18.00    | 18       | 18       | 18       |          |          |          |          |
| Breeding Years                       | 5        | 5        | 8        | 5        |          |          |          |          |
| Babies Birth <sup>-1</sup>           | 1        | 6        | 1        | 1        |          |          |          |          |
| Wean Age                             | 60       | 30       | 60       | 60       |          |          |          |          |
| Gestation                            | 150      | 115      | 279      | 150      |          |          |          |          |
| Oestrous Cycle                       | 21       | 21       | 21       | 21       |          |          |          |          |
| Lactation Time                       | 210      | 30       | 240      | 210      |          |          |          |          |
| Ratio F:M breed stock                | 20       | 20       | 20       | 20       |          |          |          |          |
| Desired Herd Size                    | 50       | 0        | 10       | 50       |          |          |          |          |
| Mixed Herd Numbers                   | 25       | 0        | 2        | 25       |          |          |          |          |
| Milk grams day <sup>-1</sup>         | 0.94     | 0        | 6        | 0.9      |          |          |          |          |
| Milk MCals Kg <sup>-1</sup>          | 0.940    | 0        | 0.94     | 0.940    |          |          |          |          |
| Ewe Fleece Kg Yr <sup>-1</sup>       | 2.25     |          |          |          |          |          |          |          |
| Ram Fleece Kg year <sup>-1</sup>     | 1.75     |          |          |          |          |          |          |          |
| Juvenile Fleece Kg Yr <sup>-1</sup>  | 1.4      |          |          |          |          |          |          |          |
| Gestation Req'd                      | 3.970    | 3.970    | 6        | 3.970    |          |          |          |          |
| Lactation Req'd                      | 5.410    | 5.410    | 7.5      | 5.410    |          |          |          |          |
| Maintenance Req'd                    | 3.200    | 0        | 5.8      | 3.2      |          |          |          |          |
| Juvenile's Req kg <sup>-1</sup>      | 0.030    | 0.030    | 0.030    | 0.030    |          |          |          |          |
| Land Energy Hectare <sup>-1</sup>    | 1500     | 1500     | 1500     | 1500     |          |          |          |          |
| Fodder Fed From Crop                 | Crop 10  | Crop 10  | Crop 10  | Crop 10  |          |          |          |          |
|                                      | Barley   | Barley   | Barley   | Barley   |          |          |          |          |
| Grams <sup>-1</sup> Straw/Chaff      | 1000     |          | 4956     | 1000     |          |          |          |          |
| Grams <sup>-1</sup> Grain            | 750      | 750      | 1500     | 750      |          |          |          |          |
| Days on supplemental feeding program | 90       | 90       | 90       | 90       |          |          |          |          |

CROP TARGETS

|                       | Percentage | MCals for   | Calorie Tgt | Percent | Percent for | Prdn Tgt | Area Target | Fallow MCals raw/Chaff MC |           |
|-----------------------|------------|-------------|-------------|---------|-------------|----------|-------------|---------------------------|-----------|
|                       | Tame Diet  | Animal Feed | Mcal        | Wastage | re-seeding  | Kg       | Ha          | Available                 | Available |
| TRUE Horse Bean       | 0.52%      | 0           | 8968        | 30%     | 20%         | 4484     | 18          | 4484                      |           |
| TRUE Einkorn          | 1.54%      | 0           | 26559       | 30%     | 20%         | 13280    | 53          | 13280                     |           |
| TRUE Emmer            | 44.11%     | 0           | 760732      | 30%     | 20%         | 380366   | 1,521       | 380366                    |           |
| TRUE Lentils          | 31.80%     | 0           | 548431      | 30%     | 20%         | 274215   | 1,828       | 457026                    |           |
| TRUE Peas             | 3.59%      | 0           | 61914       | 30%     | 20%         | 28576    | 381         | 95252                     |           |
| TRUE Olives           | 3.58%      | 0           | 61742       | 30%     | 0%          | 20066    | 33          | 16722                     |           |
| TRUE Grapes           | 8.20%      | 0           | 141419      | 30%     | 0%          | 61282    | 102         | 51068                     |           |
| FALSE Grass Peas      | 0.00%      | 0           |             | 30%     | 20%         |          |             |                           |           |
| TRUE Bitter Vetch     | 0.51%      | 0           | 8796        | 30%     | 20%         | 4398     | 22          | 5497                      |           |
| TRUE Barley           | 6.15%      | 1635221     | 1741286     | 30%     | 20%         | 870643   | 3,483       | 870643                    | 741788    |
| FALSE FreeHull Barley | 0.00%      | 0           |             | 30%     | 20%         |          |             |                           |           |
| TOTALS                | 100.00%    | 1635221     | 3359846     |         |             | 1657309  | 7,442       | 1894338                   | 741788    |

ANIMAL TARGETS

|            | Percentage | Calorie Tgt | Prdn Tgt | Total MCals | MCals From  | MCals From | MCals From  | MCals From | Ha       |
|------------|------------|-------------|----------|-------------|-------------|------------|-------------|------------|----------|
|            | Tame Diet  | Mcal        | Kg       | Feed Reqd   | Straw/Chaff | Grain      | Fallow Land | Land       | Required |
| TRUE Sheep | 12.26%     | 91,624      | 28,632   | 6,011,520   | 335,515     | 531,626    | 608,996     | 4,535,383  | 3,024    |
| TRUE Pig   | 39.62%     | 296,095     | 92,530   | 3,441,660   | 0           | 429,188    | 348,657     | 2,663,816  | 1,776    |
| TRUE Cow   | 37.74%     | 282,045     | 88,139   | 4,156,372   | 350,758     | 224,285    | 421,061     | 3,160,268  | 2,107    |
| TRUE Goat  | 10.38%     | 77,574      | 24,242   | 5,089,821   | 284,077     | 450,123    | 515,624     | 3,839,997  | 2,560    |
| 4 TOTALS   | 100.00%    | 747,338     | 233,543  | 18,699,373  | 970,350.6   | 1,635,221  | 1,894,338   | 14,199,464 | 9,466    |